

Customer Churn Prediction System

Task ID: ML-2

Domain: Machine Learning

1. Dataset Description

For this project, a realistic synthetic dataset containing 1,500 customer records was programmatically generated using the Python Faker library. The dataset reflects a subscription-based business model and includes 13 key attributes:

- **Demographics:** Customer ID, Age, Gender, City.
- **Account Information:** Subscription Type, Monthly Spending, Tenure.
- **Behavioral Metrics:** Number of Purchases, Customer Support Requests, Login Frequency, Last Activity Date, Satisfaction Score.
- **Target Variable:** Churn Status (Binary: 0 for Retained, 1 for Churned).

The dataset was embedded with real-world logical patterns (e.g., customers with low satisfaction and high support requests having a higher probability of churning) to train the models effectively.

2. Data Preparation Process

The raw data underwent several preprocessing steps to make it suitable for machine learning algorithms:

- **Data Cleaning:** Non-predictive columns such as `Customer ID` and `Last Activity Date` were dropped from the feature set to prevent the model from learning irrelevant patterns.
- **Validation:** The generated dataset was verified to ensure no missing values or duplicated records existed.
- **Train-Test Split:** The dataset was divided into a training set (80%) and a testing set (20%) using `train_test_split` with a fixed random state for reproducibility.

3. Feature Engineering

To handle categorical variables, the following encoding techniques were applied:

- **Label Encoding:** Applied to the `Gender` column, converting binary text categories into numerical format (0 and 1).
- **One-Hot Encoding:** Applied to multi-class categorical features like `City` and `Subscription Type` using `pd.get_dummies()`.

- **Feature Alignment:** During the web deployment phase, the `.reindex()` method was implemented to ensure the live input features perfectly matched the trained model's expected features, filling any missing categorical columns with zeros to prevent dimension mismatch errors.

4. Model Selection & Hyperparameter Tuning

Two powerful ensemble learning algorithms were selected for this classification problem:

1. **Random Forest Classifier**
2. **XGBoost Classifier**

To achieve maximum predictive capability, **Hyperparameter Tuning** was performed using `GridSearchCV`. Various combinations of `n_estimators`, `max_depth`, and `learning_rate` were rigorously tested.

5. Evaluation Results

The models were evaluated using multiple metrics including Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

- After iterative data refinement and hyperparameter optimization, the models successfully achieved an **accuracy exceeding 80%**.
- The **Random Forest** model consistently demonstrated robust performance in capturing the complex, non-linear relationships between customer satisfaction, support requests, and the likelihood of churning.

6. Key Findings

The exploratory data analysis and model training revealed critical insights into customer behavior:

- **Satisfaction & Support:** There is a strong correlation between low satisfaction scores (≤ 2) combined with high customer support requests (≥ 4) and immediate customer churn.
- **Early Flight Risk:** New customers (tenure < 6 months) who incur high monthly spending are at a significantly higher risk of leaving compared to long-term subscribers.
- **Engagement Indicators:** High login frequencies paired with zero support requests are the strongest indicators of customer retention and stability.

7. Business Recommendations

Based on the predictive model's insights, the following data-driven strategies are recommended to reduce churn:

- **Proactive Intervention:** The customer support team should immediately flag and reach out to customers registering more than 3 support requests, offering personalized assistance or retention discounts before they decide to leave.
- **Onboarding Optimization:** Implement a specialized onboarding and check-in process for new customers with high-tier subscriptions to ensure they realize the platform's value early on.
- **Engagement Campaigns:** Continue rewarding stable customers (high login frequency, high tenure) with loyalty perks to maintain their high satisfaction scores.